LoDA-SA: Low-Frequency Distance-Aware Sparse Attention

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***Abstract*—Sparse attention must retain useful candidates at low selection cost. We study LoDA-SA, a distance-aware two-level selector: a near–far page-to-token cascade with shared 32-coordinate low-frequency RoPE scoring, original-dimensional attention on selected tokens, and absolute budgets: protected sink and window lengths, near extent n_l, near page budget n_b with far pages $K_1 - n_b$ by subtraction, and one free-competition token pool; nonaligned protection-boundary pages join candidates outside the quotas. Ratio-coupled parameterizations instead silently reallocate far capacity: one expression links the far token budget to three knobs, and archived diagnostics exhibit all three failures, including an exact integer threshold. The absolute interface makes page capacities explicit and drops regional token quotas. An archived 86-configuration LongBench summary scores quota-free selection at 32 near pages and 4096-token extent 48.26 versus 47.46 for a no-partition control, with a numeric-token-quota arm at the same geometry reaching 48.67; these single-run results isolate neither regional axis nor quota removal. Single measured runs of the compact design score 59.02, 47.80 and 46.82 at RULER 32K/64K/128K against separate-run dense anchors of 59.38, 48.54 and 46.43, gaps at the scale of historical batch-level spread, while a historical ratio-interface arm scores 23.98 at 128K with zero far tokens. These runs establish neither non-inferiority nor mechanisms; graph replay with the fused build undercuts dense TPOT at all measured lengths (1.45/2.22/3.64× at 32K/64K/128K), and the compact 4/128 design still requires matched quality and preparation-inclusive complete-request cost.**

***Index Terms*—sparse attention, shared representation, regional retrieval, long-context inference**

I. Introduction

Long-context decoding must retain access to useful history under a finite attention budget. Tokens at different distances compete for this budget, and global ranking alone does not reserve candidate access for a positional region. Distance therefore offers a way to organize selection, without assuming that importance decreases monotonically with distance or that near tokens are always preferable. A selector must decide which regions can supply candidates and then rank those candidates cheaply enough to justify restricting final attention.

Token-level selection tracks fine-grained relevance but pays a per-token indexing cost; block-level summaries cut that cost but blur where attention concentrates within a page [4]. Following ClusterKV's token/page perspective [8], Figure 1(a) shows how each summary fails in its own way on a single Qwen3 query: mean-key pooling dilutes concentrated far spikes and misses 8 reference pages holding 86.5% of reference mass, while min–max extrema miss more pages (9) carrying far less mass (4.7%), mostly near, yet admit outside-reference pages through loose bounds. Each mechanism also ships its own scoring view, so cascades recompute coordinates and summaries that one shared set could serve. Unequal regional allocations and realized counts preclude a controlled comparison but motivate examining how scoring and budgets shape candidate access.

The complementary failure is systematic, not an artifact of one query: across 128 correlated observations from four sequences, avg misses are far-heavy in 65% of queries and minmax misses near-heavy in 61%, with mean regional miss shares .084 near/.170 far for avg and .079/.042 for minmax (S6). The complementarity establishes neither regional specialization nor near-region equivalence, and with a fixed 2048-token cap the best of 23 tested near-avg/far-extrema splits has mean missed mass .135 versus .121 for global extrema: a bounded scan that establishes neither a hybrid advantage nor a task-quality benefit.

We study a two-level selector that couples a shared scoring view with absolute regional budgets. Both stages use a fixed set of 32 coordinates comprising 16 complete low-frequency RoPE pairs; the near page budget n_b and near extent n_l are absolute, far pages follow by subtraction, nonaligned protection-boundary pages join candidates outside the quotas, and L2 refines tokens in a single free-competition pool. Among the three 32-coordinate subsets in Figure 1(b), low-frequency pairs retain .84 of full-dimensional reference mid-TopK mass on average, random coordinates .51, high-frequency pairs .24; the random subset does not preserve RoPE pairs. Selected tokens use original-dimensional Q/K/V in one final attention operation. Compact coordinates narrow each scoring dot product, but ordinary L2 still scores the full history before masking; fewer shortlisted pages do not remove that work. We also formulate a prefill-informed far gate motivated by layer variation in Figure 1(c); it is disabled by default.

We contribute LoDA-SA, an absolute-budget two-level selector with shared low-frequency coordinates, a unified analysis of implicit budget coupling in ratio parameterizations, a descriptive grid comparison of regional geometry and token allocation, and separate evaluations of task quality, regional coverage and execution cost. Under the archived ratio interface, one expression in the page cap, near share and fill fraction determines the far token budget, so raising any of the three can zero it; reported Qwen3-8B RULER 32K AVG moves 0.99 points when archived near quotas shift (16,640) to (8,320), with both quotas changing together. Grid cells are single runs; no-graph sparse decoding runs slower than dense at every measured length, and establishing a quality–latency benefit requires matched quality and preparation-inclusive complete-request measurements.

II. Related Work

Quest supplies sign-aware page-extremum bounds, SparQ selects key coordinates from queries, and HISA prunes the attention set hierarchically [1], [2], [3]. AsyncTLS explicitly follows Quest for coarse block selection and applies Double Sparsity for fine token selection; its cascade also includes quantization, per-query normalization with GQA probability aggregation and original-KV attention [4]. Earlier Twilight uses a select-then-prune architecture [14]; LoDA-SA studies explicit distance-region page access within this hierarchical-selection paradigm, with complete low-frequency RoPE pairs shared across both stages and unified L2 competition. The relationship is algorithmic; code lineage is unverified.

The selector's own reads also matter. Token-Picker uses conservative probability estimates from partial key-bit chunks to stop further key-chunk and value transfers for pruned tokens [13]; our selector instead ranks candidates with a fixed coordinate subset under page budgets, keeping original-dimensional KV. Our selector keeps both selection stages in bf16: AsyncTLS token keys are INT4, FFD scans 2-bit keys, and official HISA pools block summaries in FP8 [4], [5], [3]; LoDA-SA targets lower selection overhead through compact scoring and structured candidate admission, not numeric compression, and ordinary L2 still scores full history, so narrowing alone establishes no saving. KV compression and placement methods

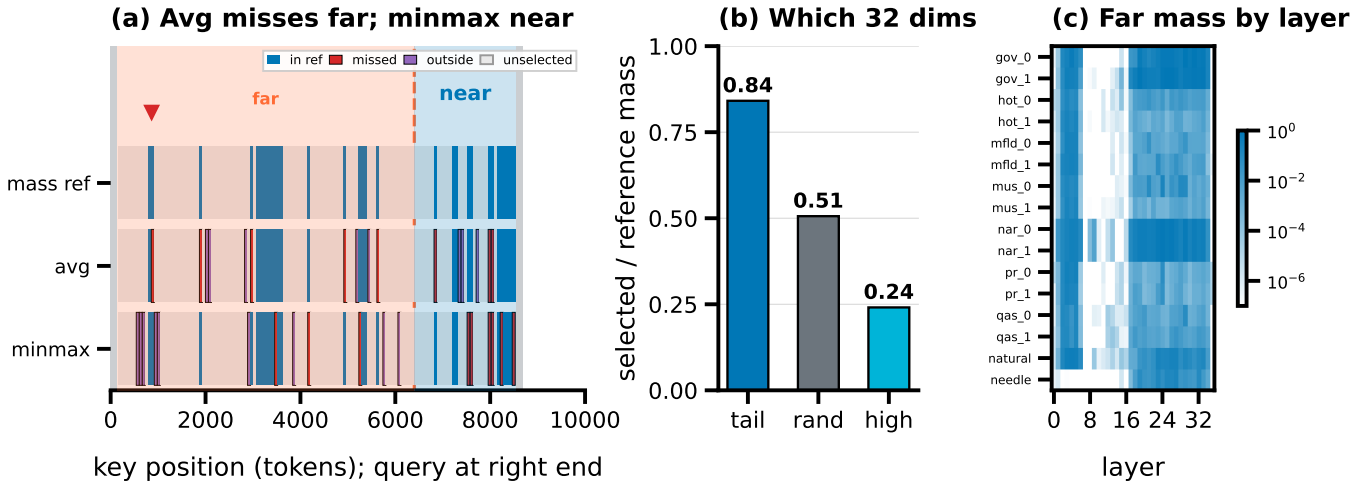


Fig. 1. Motivating diagnostics for LoDA-SA, one panel per design decision; Qwen3-8B, 128/128 prefix/recent protection in (a) and (b). (a) HotpotQA sample 0, layer 24, query 8672, an extreme-complementarity pick (far share 87th percentile of 128): strips for mass-ranked reference, avg and minmax selections (2017/2017/2048 tokens). Avg misses 8 reference pages holding 86.5% of reference mass, three quarters far; minmax misses 9 pages holding 4.7%, four fifths near, recovering 6 of avg's 8 misses (all five far); the marked spike page holds 99.85% of its mass on one token that avg dilutes and minmax retains. Over 128 correlated observations from four sequences, avg misses are far-heavy in 65% of queries and minmax near-heavy in 61% with regional miss shares .084 near/.170 far for avg, .079/.042 for minmax (S6). Blue: selected reference; red: missed reference; purple: selected outside. Mass-rank is a whole-page reference, not an optimal oracle, TopK or DSA; unequal allocations and counts preclude specialization or task-quality claims. (b) Selected mass relative to full-dimensional reference mid-TopK mass: tail 32 coordinates retain .84, random32 .51, high32 .24; random32 does not preserve RoPE pairs. (c) A 16-by-36 historical proxy matrix: 24/36 layer medians below .01, 12–32 low-proxy layers per sequence (last sampled query, group-logit summation, far interval $[64, t-2048)$; S6), unlike the gate statistic. These are diagnostics, not task-quality or decode-safety results.

address complementary storage and movement constraints [6], [7], [8]; a smaller scoring view retains original-KV capacity without establishing lower complete-request latency.

III. LoDA-SA: Distance-Aware Selection in a Shared Subspace

LoDA-SA separates three decisions: which distance regions can admit pages, how to score them with one shared compact view, and which pooled tokens enter final attention. Fig. 1 motivates the first two directly: complementary near/far misses drive regional admission (a), and the frequency contrast drives the shared view (b); panel (c) motivates an optional far gate (Sec. III-F). The distance policy uses n_l for near extent and n_b for its page budget, leaving $K_1 - n_b$ far pages. The implementation prepares the shared view before regional L1 selection (Fig. 2); selection runs only during decoding with dense prefill, gating stays disabled, and archived runs do not validate current LoDA-SA.

A. Regional budgets admit pages from both bands

In decoding, t indexes the current token across the full sequence including prefill. Let s and w denote the protected sink and recent-window lengths, with protected set $\mathcal{P}_t = \{i : 0 \le i \le t,\ i < s \text{ or } i > t - w\}$ and unprotected middle $\mathcal{M}_t = \{0, \dots, t\} \setminus \mathcal{P}_t$; overlapping protection counts once. The requested near length is n_l tokens. The implementation uses at least one page, $L = \max(B, n_l)$, then floors the near/far cut to a page boundary: near ends at the protected window, and far is the remaining middle history.

Selected near and far pages, together with protection-boundary bypass pages, form one L2 candidate pool. There is no prescribed near/far token split: regional representation in the final set follows the scores, not a token quota. With enough positive-probability candidates, L2 cannot recover unprotected tokens from L1-rejected pages. S7 details scarce-support and masked zero-tie exceptions.

B. One 32-coordinate view scores both stages

Both regions and both selection stages use the same coordinate map within each layer and KV head, while each query head is scored separately. For the Qwen3-8B tail-coordinate configuration, we select post-RoPE coordinates $J = \{48, \dots, 63, 112, \dots, 127\}$ from each 128-dimensional query and key. The indices retain 16 complete low-frequency split-half RoPE pairs (Supplementary Fig. S1); this statement does not describe generic full-width, static-pair, projection or MLA configurations. With coordinate-selection matrix $E_J \in \mathbb{R}^{128\times 32}$, the scoring vectors are $z_q = E_J^\top q$ and $z_i = E_J^\top k_i$. The retained pairs' slow positional rotation motivates this selection, although it does not guarantee preservation of important tokens. The compact implementation uses width-32 scoring tensors (S7 details the historical width-128 form).

L1 aggregates selected key coordinates into page summaries for coarse ranking; L2 refines candidates with per-token scores in the same coordinate view. Original KV remain full width. Compact scoring narrows each dot product fourfold, but view preparation and maintenance add work without compressing the original KV cache (S7).

C. Sign-aware bounds and boundary bypass

Regional page caps and one shared token budget constrain the cascade (Supplementary Fig. S2). Near pages use query–mean-key dot products to summarize page-level relevance; far pages use sign-aware extrema to bound shared-space token scores, a heuristic assignment. Summaries cover full physical pages, including protected positions (S7). In the partitioned path, a page containing a non-page-aligned sink end or recent-window start bypasses L1 ranking: it joins outside the page quotas, and its unprotected tokens still compete in L2; only protected tokens are forced into final support. For a nonempty page b, let $m_{b,j} = \min_{i\in b} z_{i,j}$ and $M_{b,j} = \max_{i\in b} z_{i,j}$. Quest's

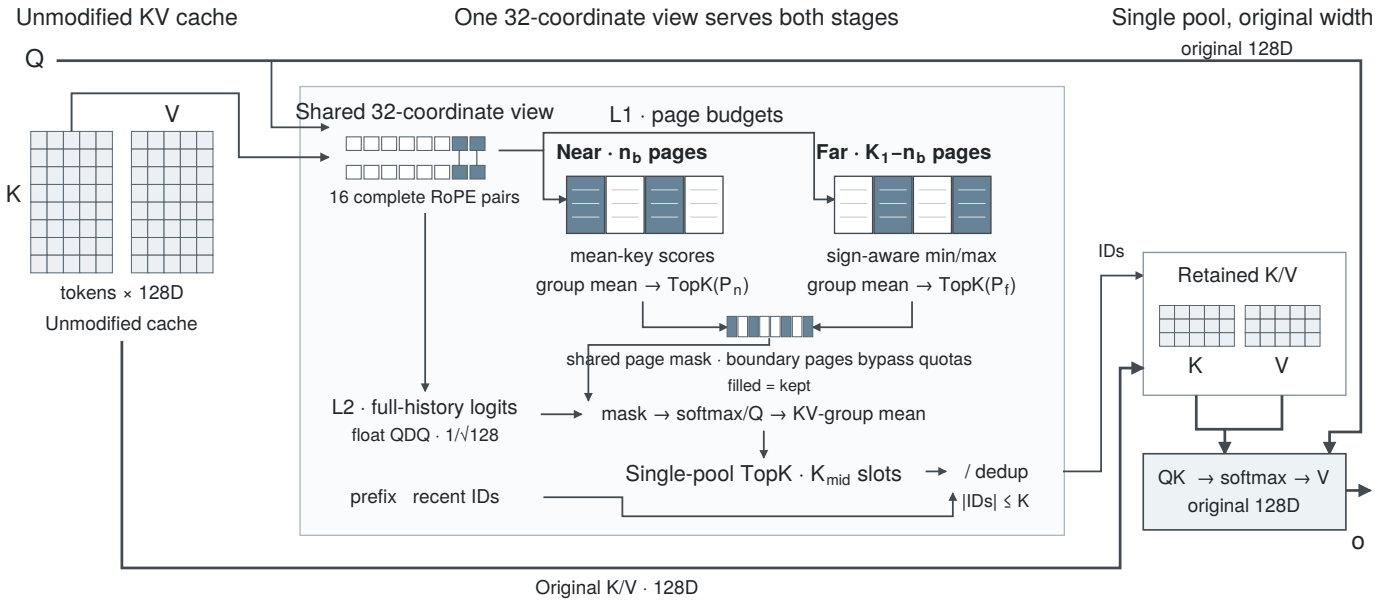


Fig. 2. LoDA-SA reference architecture: boundary bypass enabled, gating disabled. One shared coordinate view feeds regional L1 page scoring and ordinary full-history L2 scoring; lanes carry absolute budgets (n_b near pages, $K_1 - n_b$ far pages, K_{mid} pooled slots). Nonaligned protection-boundary pages join the page mask outside L1 quotas; only protected tokens bypass final selection. Per-query softmax and KV-group probability averaging precede one pooled middle-token TopK; protected positions join deduplicated IDs within K. IDs define support over original-width K/V; physical gathering is backend-dependent. The optional far gate (prefill-calibrated, frozen at decode) is disabled. Arrays are schematic.

construction [1] gives

$$U_b(z_q) = \sum_j (z^+_{q,j} M_{b,j} + z^-_{q,j} m_{b,j}), \tag{1}$$

where $z^+ = \max(z, 0)$ and $z^- = \min(z, 0)$. Each signed coordinate contribution is bounded by its chosen extremum, so $z_q^\top z_i \le U_b(z_q)$ for every shared-space key in b. This does not bound original-dimensional logits. Extrema may come from different tokens, making the bound loose enough to misrank pages and miss important tokens; average-key scores lack this per-token bound.

D. Absolute budgets expose what ratio coupling hides

For total cap $K \ge |\mathcal{P}_t|$ and disjoint protection, the middle-token budget $K_{\text{mid}} = K - |\mathcal{P}_t|$ fills one pool: L2 applies the retained-page mask, then a single TopK over group-averaged probabilities retains up to K_{mid} tokens regardless of region. The interface exposes absolute quantities: near page budget n_b, far pages $K_1 - n_b$ by subtraction (page quotas clipped to availability), and near extent n_l. The far-page budget still depends jointly on K_1 and n_b; explicit parameters do not make regional capacities independent.

Ratio-coupled interfaces hide this subtraction. The archived parameterization derives $n_b = \max(1, \text{round}(\beta K_1))$ and $n_k = \min(\lfloor \gamma n_b B \rfloor, K_{\text{mid}})$ with ties-to-even rounding, so the far token budget reduces to

$$k_f = \max(0,\, K_{\text{mid}} - \min(\lfloor \gamma n_b B \rfloor,\, K_{\text{mid}})). \tag{2}$$

Three knobs, K_1, β and γ, therefore reallocate far capacity through one expression, while far pages remain admitted at $k_f = 0$: page eligibility and token capacity are separate constraints. Archived diagnostics exhibit all three reallocations, including an exact integer threshold at which raising K_1 zeroes far tokens (Q2, S6). The current interface removes the prescribed regional token split, but does not guarantee regional access: $n_b = K_1$ leaves no far-page quota, and pooled L2 can select no far tokens even when far pages are admitted.

With sufficient history and available candidate pages, an illustrative compact configuration ($K = 1024$, $s = 4$, $w = 128$, $n_b = 32$, $n_l = 4096$, $K_1 = 64$) leaves 892 middle slots per KV head, with L1 admitting 32 near and 32 far pages; the partial prefix page can reselect up to four protected tokens, so retained coverage is nominal.

E. One merged softmax in original dimensions

GQA assigns four of Qwen3's 32 query heads to each of its eight KV heads. L1 averages their page scores before regional TopK. Ordinary L2 computes each query's full-history logits, applies the shared page mask, then softmax. Group-averaged probabilities determine one TopK across the pooled middle tokens. L2 uses scale $1/\sqrt{128}$ before softmax even with 32 scoring coordinates. Because softmax precedes group averaging, this rule generally differs from applying softmax to group-averaged logits.

For KV group h, let $\mathcal{S}_{t,h}$ be the deduplicated union of protected positions and pooled winners, with $|\mathcal{S}_{t,h}| \le K$. Each query g uses this support for its own full-dimensional attention ($d = 128$):

$$o_{h,g} = \sum_{i \in \mathcal{S}_{t,h}} \frac{e^{q_{h,g}^\top k_{i,h}/\sqrt{d}}}{\sum_{j \in \mathcal{S}_{t,h}} e^{q_{h,g}^\top k_{j,h}/\sqrt{d}}} v_{i,h}. \tag{3}$$

Each query applies one softmax over the merged indices, avoiding summed separately normalized regional outputs [9]. Restricted denominators can still change the output; physical gathering is backend-dependent, and short sequences can use a dense masked path.

F. Optional far gating stays disabled

The current n_l/n_b design keeps far gating disabled. A historical extension calibrates far mass from final prefill queries after per-head causal softmax, then freezes a layer decision at sparse decoding; its

TABLE I
Compact design parameters, matching the measured decode runs in Q3; K includes unique protected tokens. The measured RULER quality arms (Q1/Q2) predate the halved L1 cap and run K_1=128; a five-task LongBench pair scores 48.10 at K_1=64 versus 47.91 at K_1=128 (single runs, S6).

Parameter	Setting
Final token cap K	1024
Page size; L1 page cap K_1	64; 64
Near page budget n_b (far)	32 (32)
Near extent n_l	4096 tokens, page-aligned
L2 token selection	single pool, free competition
L1/L2 active coordinates	32/32
Prefix / recent protection	4/128

numeric-token-quota refill branch is not the current unified L2 rule (S5). In the unified branch, removing far-page competition does not remove all final far tokens: a sink-boundary page may still enter the pooled competition, so we claim no complete far skipping and transfer no historical gate results. Calibration thresholds do not bound decode error; quality and complete-request cost require separate evaluation.

IV. Experimental Setting

We evaluate task quality (Q1), regional coverage (Q2) and execution cost (Q3) separately. Table I defines the compact Qwen3-8B design: dense prefill, sparse decoding, a 1024-token cap with four prefix and 128 recent tokens (4/128 protection). The Q2 grid supplies historical descriptive comparisons and motivates the compact operating point $(n_b, n_l) = (32, 4096)$, its best quota-free cell, as a descriptive choice; filling K_{mid} by free competition remains a design choice, not evidence that deleting a numeric token quota preserves quality. Archived runs retain ratio-defined geometries and quotas; $(8, 320)$ denotes near pages and tokens, not (n_l, n_b). Their quality runs applied 16-level quantize–dequantize L2 token keys outside the mainline and, for the shared-subspace arm, 128/128 protection with 32 active coordinates in zero-filled width-128 L2 keys (S7). The labels "full-dimensional page/token scoring" denote reported variants without complete execution configurations, not controlled ablations.

Workloads and comparisons. Reported quality covers five-task LongBench [10] and 11-task RULER [11] (Qwen3-8B: 32K/64K/128K), plus ported Quest/AsyncTLS/JSSA/FASA baselines. FullKV is dense and uncapped (RULER/decode timing). Method comparisons must disclose native prefill/decode paths; policy-matched adaptations need separate reporting. Budget matching aligns final unique-token caps K including protection; realized counts and head sharing affect KV traffic, and equal K alone establishes neither quality nor cost matching (S7).

Metrics. Primary quality is task-score AVG over each fixed benchmark/length set; per-task gains and losses explain AVG without zero-loss requirements. Policy-matched controls require matched models, samples, lengths, scorers, phase policies and aligned sparse-token budgets; incomplete sets lack full-set AVG. Regional miss shares normalize each query's per-region missed reference mass by its mass-ranked reference total, then average across correlated observations (S6); panel (b) normalizes by full-dimensional mid-TopK mass. Neither selector time (synthetic single-layer) nor synchronized TPOT (model-level post-first-output intervals) measures preparation-inclusive complete-request latency, so unpaired studies define no quality–latency frontier; AVG differences are descriptive, not non-inferiority evidence.

V. Evaluation and Discussion

A. Q1: Does reduced scoring width preserve task quality?

On HotpotQA, Qasper, MuSiQue, GovReport and RepoBench-P [10], five-task means fall within a 0.33-point spread: shared-subspace 47.588, the nominal page variant 47.910 and the token variant 47.666 (Supplementary Fig. S4). These differences describe recorded configurations, not evidence that reduced width preserves quality: the variants are not verified single-factor interventions. All three arms are sparse; dense FullKV is a different reference.

B. Q2: How do regional budgets constrain candidate access?

Descriptive grid comparisons. An archived Qwen3-8B summary reports 86 near-page, near-extent and token-quota configurations plus one no-partition control on five LongBench tasks, with one run per configuration. Reported AVG5 is 48.264 for quota-free selection at $(n_b, n_l) = (32, 4096)$, 48.300 for the best quota-free configuration and 47.462 for the control; these are descriptive point estimates. Near-page marginal means rise from 46.37 (n_b=4) to 48.21 (n_b=32) and flatten at 64 (48.11), and best cells cluster near $n_b B \approx n_l/2$, matching archived near-region mass concentration (top-half pages hold a median .93); these single-run marginals support no causal isolation. Across 69 numeric-quota versus quota-free comparisons over 17 distinct geometries, the mean AVG5 difference is $+0.040$ points (42 positive). At $(n_b, n_l) = (32, 4096)$, the numeric-quota-448 arm scores 48.668 versus 48.264 without a quota, a $+0.404$-point difference. The three paired intervals compare top-three numeric-quota configurations with one another, not with quota-free selection, so containing zero establishes neither equivalence nor absence of a token-quota effect. These summaries isolate no axis's quality contribution and do not validate the pooled-L2 configuration; matched identities, paired predictions and complete-request costs remain necessary (S6).

Archived RULER 32K arms move 58.30→59.29 as near quotas shift $(16, 640) \rightarrow (8, 320)$ (archived $\beta = .125 \rightarrow .0625$), with far pages rising 112→120 and the far token budget 252→572 jointly; the near-page-4/8 arms score 58.07 ($K_1 = 128$, far tokens 732) and 58.96 ($K_1 = 256$, near quotas $(8, 320)$, far pages 248), both below 59.29, the latter differing by CWE $+3.9$ and FWE -9.67. Records match; predictions are unreplayed.

Implicit coupling failures. Archived diagnostics exhibit all three reallocations of Eq. (2). Integer enumeration of the archived formulas locates the first far-zero page cap at $K_1 = 156$ for $K = 1024$ ($B = 64$, $\beta = .125$, $\gamma = .625$, 128/128 protection); far budget 8 remains at $K_1 = 155$, so capacity appears to grow while far tokens vanish. Archived task scans collapse HotpotQA from 53.6–57.3 to 37.7 exactly on zero-far-budget arms, graded by far fraction (16.4 points at .875, about 1 at .125). At the archived grid cell $(\alpha, \beta) = (.25, .125)$, free competition ties the best γ within .001 in retained-mass diagnostics (.852 versus .8515), while $\gamma = 0$ reads as far-free but zeroes the near token quota, dropping retained mass to .827. The historical 128K RULER arm computed $k_f = 0$ under its reported budgets (S6). These archived diagnostics unify the three anomalies but are not new controlled experiments.

Reported 32K AVG gap to dense FullKV is -0.09 across machines; at 64K, AVGs are 48.85/48.05 for near quotas (8,320)/(16,640) versus 48.54. Sparse arms share a machine and scorer hashes but differ in prediction scripts; the dense anchor differs in scorer and identity, and missing paired intervals preclude causal conclusions (S8).

Ported baselines. Author-reported RULER32K AVGs: 56.00 for a historical numeric-token-quota arm at nominal $K = 512$ versus

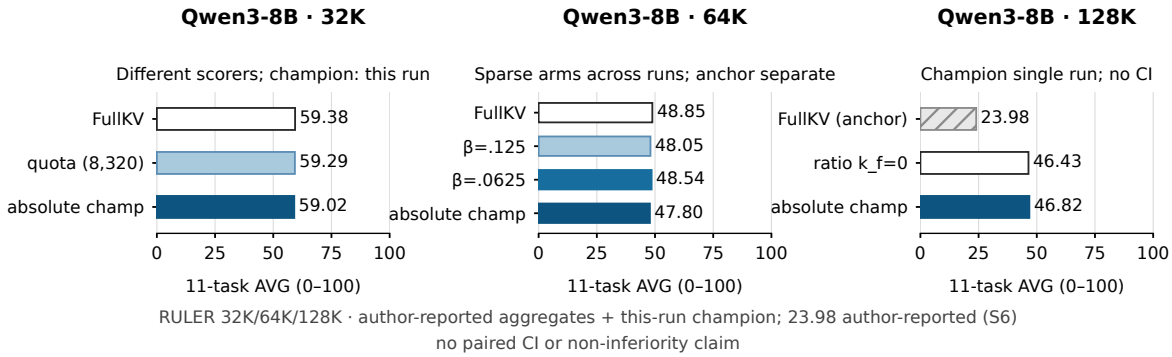


Fig. 3. RULER AVG, Qwen3-8B, 11 tasks, 100 examples/task/arm. Archived sparse arms report $K = 1024$, 4/128 protection and ratio-derived quotas, predating free competition; 32K near quota (8,320) ($\beta = .0625$), 64K shows (8,320) and (16,640). The measured arm runs Table I's compact design once per length (free competition, batch 8/2/2, YaRN off/2.0/4.0). Dense anchors are separate runs; 32K spans machines and scorers. The hatched 128K bar is the historical ratio interface with $k_f = 0$, scoring 23.98 versus 46.82 for the measured absolute interface. Single runs without paired intervals keep all comparisons descriptive (S8).

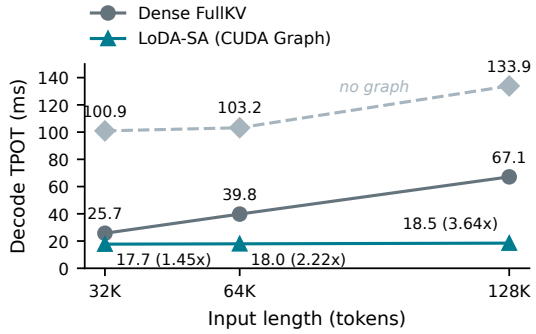


Fig. 4. Measured decode TPOT, Qwen3-8B, Table I geometry with the fused kernel chain: dense FullKV (native decode) versus LoDA-SA with whole-step CUDA Graph replay; the dashed arm disables it. 127 after-first intervals include synchronization and argmax (dense 6 runs, graph 4, eager gate arm). Graph replays are internally bitwise identical; against eager decoding 64K is bitwise identical, and 32K and 128K each show one single-token divergence (bf16 tie-breaks in the enlarged far sort at K_1=64). Intervals measure no preparation-inclusive complete-request latency and no matched quality–cost comparison (S6).

48.33 for the AsyncTLS port, a 7.67-point gap; historical arms at $K = 1024$ and $K = 384$ score 59.29 and 15.00, with port scores 58.7 (JSSA), 42.85 (Quest) and 18.64 (FASA). Budgets, protection and phases are unverified as matched, and these arms do not validate the current pooled-L2 configuration; S8 gives provenance, the CWE loss of 43.4 points and the missing paired intervals.

At 128K, the (8,320)/(16,640) arms score 45.73/46.02 on one machine; 45.73 is 98.49% of the separate-run dense anchor 46.43. Historical RULER 128K yields 23.98 under budgets implying $k_f = 0$, execution unverified, the 22.45-point loss unexplained (S6). Five-task LongBench means are 26.09 (protection-only, 256 tokens) versus 47.91 (reused sparse, 1024); retrieval and budget both change.

Measured compact-design runs. Single runs of the pre-halving Table I configuration (K_1=128; free competition; batch sizes 8/2/2, YaRN off/2.0/4.0) score 59.02, 47.80 and 46.82 at RULER 32K/64K/128K, 11 tasks at 100 examples each, against separate-run dense anchors of 59.38, 48.54 and 46.43: gaps of -0.36, -0.74 and $+0.39$, the scale of historical batch-level differences (same-machine 64K sparse arms differ by 0.80; the cross-machine 32K comparison by 0.09). The anchors span generations, machines and scorers, and single runs carry no paired intervals, so these comparisons establish neither non-inferiority nor degradation. At 128K the measured absolute interface scores 46.82 where the historical ratio arm with $k_f = 0$ scored 23.98; without matched protocols, that gap remains descriptive.

C. Q3: Does compact scoring reduce execution cost?

Model-level decoding. With whole-step CUDA Graph replay, the fused build decodes at 17.7/18.0/18.5 ms at 32K/64K/128K versus dense 25.7/39.8/67.1 ms: 1.45/2.22/3.64×, monotone in length, with 32K already a 1.45× gain rather than parity (Fig. 4). Sparse cost grows 4.2% across the 4× length sweep while dense grows 161%. The same build replicates on Llama-3.1-8B: 15.0/15.2/15.6 ms versus dense eager 21.4/33.3/56.8 ms (1.43/2.19/3.64×), so the length-monotone gain is not Qwen-specific. Disabling the graph returns the sparse arm to 101–134 ms: the gain tracks replay, not selection. Timings include synchronization and argmax and measure no preparation-inclusive complete-request latency (S6).

Kernel-fusion build. Relative to the pre-fusion build of Fig. 4's protocol (24.7/27.4/32.4 ms), six fused Triton kernels cut the same geometry at K_1=128 to 18.7/18.9/19.4 ms: 1.41/2.10/3.46× (single H20 GPU, medians of three synchronized runs per length; a re-run verify cell reproduces the prior judgment at 18.668 versus 18.675 ms, bit-level). Halving the L1 page cap to K_1=64 halves coarse-page work and sits at the archived down-scan quality peak (48.10 versus 47.91 on the five-task LongBench pair, single runs), giving the measured 17.7/18.0/18.5 ms above. The kernels cover mask assembly, candidate compression, summary maintenance, L1 gating and candidate assembly. At K_1=128, generated tokens stay bit-identical to the pre-fusion build across all after-first intervals of the three-length runs, and a 120-sample RULER gate (three tasks) reproduces the pre-fusion outputs byte-identically, so the fusion chain changes execution cost while leaving selection unchanged; at K_1=64 the enlarged far candidate set admits one single-token bf16 tie-break divergence between graph replay and eager decoding at two of three lengths, while graph replays stay internally bitwise identical (S6). A per-class kernel-time census at K_1=128 accounts for the near-flat length dependence (18.7→19.4 ms across a 4× context sweep). Weight GEMM (4.49/4.50/4.52 ms), selected-token attention (0.82 ms) and selected-KV gather (6.7–6.8 ms) stay constant, and summary-index maintenance is the only materially growing class (6.5→7.2

ms). Kernel busy time covers 99.7–99.9% of TPOT, leaving near-zero launch exposure. This decomposition is consistent with constant two-level budgets driving the decoupling of decode cost from length; the census itself does not isolate the mechanism.

Batch envelope. At 128K, batched decode reaches 82.6 ms per step at batch 2 versus 120.8 ms dense ($1.46\times$, 41.3 ms per sparse row). Batch 4 exhausts the 141-GB device for both arms (single H20, medians of three runs, synchronization caliber as Fig. 4). At 32K, dense overtakes at batch 4 (66.5 versus 87.1 ms) as weight GEMM amortizes across rows. The sparse batch path executes rows serially: per-row cost stays at 18.6–22.5 ms over the 32K batch sweep (batch 1–16, throughput 44–54 tokens/s) and rises to 41.3 ms at 128K batch 2. Cross-row kernel fusion remains future work. Batch flips the ordering only at 32K; at 128K the ordering survives batch 2 and ends only at the memory cap. Dense per-row cost rises 26.1→67.3 ms over the same batch runs (batch-1 points) while sparse per-row cost stays flat, a two-point pattern consistent with length, rather than batch, separating the arms. All batch rows replay one shared prompt, so these calibers measure timing only.

Ported-baseline decode cost. Eager decode of the same ports (Q2) costs 33.3/39.7/61.7 ms (JSSA), 75.6/133.0/265.9 ms (FASA), 105.3/123.4/160.3 ms (AsyncTLS) and 177.0/317.4/621.4 ms (Quest) at 32K/64K/128K, against a dense anchor of 26.1/39.6/66.9 ms in the same runs. Every port decodes slower than dense at all three lengths except JSSA at 128K (61.7 versus 66.9 ms, 8% faster), and no ported implementation ships a graph path. LoDA-SA is the only arm whose decode undercuts dense at every measured length, through whole-step graph replay plus the kernel-fusion chain above. These ports are semantic-equivalence eager implementations, so they bound port implementation cost rather than method cost (S6).

Local selector cost. A separate two-round synthetic experiment (single layer, batch one, one query, 64K QKV) reports steady-state summaries of 1.3502, 1.3598 and 1.4634 ms for nominal full-dimensional page scoring, shared-subspace scoring and nominal full-dimensional token scoring; shared-subspace minus page timing is $+0.0237$ ms in one round and -0.0046 ms in the other, establishing no consistent advantage. Missing raw sequences and an unresolved sampling discrepancy prevent uncertainty assessment; these summaries establish neither equivalence nor bottleneck attribution and are unpaired with quality runs (S6).

D. Optional gate: prefill is not a decode guarantee

Low prefill attention does not establish safe decode-time skipping. An archived input–layer record pairs prefill far-proxy below .05 with decode-tail mean .1678 over five queries: a proxy shift, not measured quality loss. Gate thresholds .05 and .10 yield five-task AVGs of 47.876 and 47.692 against the reused 47.910 reference (three tasks decline per arm; HotpotQA 1.05 and 1.26). Zero-containing bootstrap intervals do not establish non-inferiority; missing configurations, paired predictions and matched costs leave gated quality and net savings unverified (S6). The gate's value, if any, is a memory bound rather than a latency gain: a gated layer's attention domain shrinks to protection, near extent and window (4228 of 131,072 tokens at 128K), so the KV that a skipped layer *requires* falls to 3.2% of a full layer, and at threshold .05 the calibrated set covers five of 36 layers — a design-implied model-level bound near 13% that the current implementation does not materialize, since skipped-layer KV remains resident and only masked.

Design implications and limitations. Two constraints recur: ordinary L2 still scores full history, and the single pool decides final coverage by group-mean probability.

VI. Conclusion

We studied LoDA-SA: shared low-frequency page/token scoring, explicit regional page budgets, one pooled token competition, and original-dimensional attention. The archived grid reports quota-free AVG5 48.264 versus control 47.462 with a numeric-quota arm at 48.668, isolating neither quality contributions nor the compact design. Measured single runs of the compact design (59.02/47.80/46.82 at 32K/64K/128K) sit within historical batch-level spread of separate-run dense anchors, replacing the 23.98 ratio-interface loss at 128K without supporting non-inferiority claims. Graph-replayed TPOT undercuts dense at every measured length: 1.45/2.22/3.64× with the fused $K_1{=}64$ build, replicating on Llama-3.1-8B at 1.43/2.19/3.64×, with decode outputs bit-identical to the pre-fusion build at $K_1{=}128$ and graph replays internally bitwise identical throughout. A per-class kernel census ties the near-flat cost to constant two-level budgets. Batched 128K decode stays ahead of dense until the memory cap, while 32K batch 4 flips the ordering, and no ported baseline undercuts dense at all three lengths. No-graph decoding stays slower and complete-request cost remains unmeasured, so benefit still requires matched quality.

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References

[1] J. Tang, Y. Zhao, K. Zhu, G. Xiao, B. Kasikci, and S. Han, "Quest: Query-Aware Sparsity for Efficient Long-Context LLM Inference," in *ICML*, 2024. https://arxiv.org/abs/2406.10774.

[2] L. Ribar, I. Chelombiev, L. Hudlass-Galley, C. Blake, C. Luschi, and D. Orr, "SparQ Attention: Bandwidth-Efficient LLM Inference," in *ICML*, 2024, pp. 42558–42583. https://proceedings.mlr.press/v235/ribar24a.html.

[3] Y. Xu *et al.*, "HISA: Efficient Hierarchical Indexing for Fine-Grained Sparse Attention," in *COLM*, 2026. https://arxiv.org/abs/2603.28458v4.

[4] Hu *et al.*, "AsyncTLS: Efficient Generative LLM Inference with Asynchronous Two-level Sparse Attention," arXiv:2604.07815v1, Apr. 9, 2026. https://arxiv.org/abs/2604.07815v1.

[5] Z. Liu, Z. Ning, R. Li, X. Liu, Y. Song, M. Zhang, Z. He, and X. Qiu, "Faster Than Flash: Exploiting Attention Sparsity for Efficient Long-Context Decoding," in *ICML*, 2026, pp. 77913–77928. https://proceedings.mlr.press/v306/liu26da.html.

[6] H. Sun *et al.*, "ShadowKV: KV Cache in Shadows for High-Throughput Long-Context LLM Inference," in *ICML*, 2025, pp. 57355–57373. https://proceedings.mlr.press/v267/sun25b.html.

[7] Z. Wang et al., "MILLION: MasterIng Long-Context LLM Inference Via Outlier-Immunized KV Product QuaNtization," in *2025 62nd ACM/IEEE Design Automation Conference (DAC)*, 2025, pp. 1–7, doi: 10.1109/DAC63849.2025.11132862.

[8] G. Liu, C. Li, J. Zhao, C. Zhang, and M. Guo, "ClusterKV: Manipulating LLM KV Cache in Semantic Space for Recallable Compression," in *DAC*, 2025. https://doi.org/10.1109/DAC63849.2025.11132479.

[9] G. Shen, J. Zhao, Q. Chen, J. Leng, C. Li, and M. Guo, "SALO: An Efficient Spatial Accelerator Enabling Hybrid Sparse Attention Mechanisms for Long Sequences," in *DAC*, 2022, pp. 571–576. https://doi.org/10.1145/3489517.3530504.

[10] Y. Bai *et al.*, "LongBench: A Bilingual, Multitask Benchmark for Long Context Understanding," in *ACL*, 2024, pp. 3119–3137. https://aclanthology.org/2024.acl-long.172/.

[11] C.-P. Hsieh *et al.*, "RULER: What's the Real Context Size of Your Long-Context Language Models?," in *COLM*, 2024. https://arxiv.org/abs/2404.06654.

[12] G. Xiao, Y. Tian, B. Chen, S. Han, and M. Lewis, "Efficient Streaming Language Models with Attention Sinks," in *ICLR*, 2024. https://arxiv.org/abs/2309.17453v4.

[13] J. Park, M. Kang, Y. Han, Y.-G. Kim, J. Shin, and L.-S. Kim, "Token-Picker: Accelerating Attention in Text Generation with Minimized Memory Transfer via Probability Estimation," in *DAC*, 2024, doi: 10.1145/3649329.3655953.

[14] C. Lin *et al.*, "Twilight: Adaptive Attention Sparsity with Hierarchical Top-p Pruning," arXiv:2502.02770v1, Feb. 4, 2025. https://arxiv.org/abs/2502.02770v1.